

3D Cross-hole resistivity tomography to monitor water percolation during irrigation on cracking soil

A. K. Greve^{A,B}, R. I. Acworth^A, and B. F. J. Kelly^A

^AConnected Waters Initiative, University of New South Wales, affiliated with the National Centre for Groundwater Research and Training, The University of New South Wales, Sydney, NSW 2052, Australia.

^BCorresponding author. Email: a.greve@wrl.unsw.edu.au

Abstract. Irrigation water entering soil cracks can quickly move past the root-zone without being utilised by plants. To assess the efficiency of irrigation practices, reliable methods to monitor water percolation are needed. Three-dimensional (3D) cross-hole electrical resistivity tomography (ERT) was carried out during three irrigation events on soil with different surface-crack intensities. Changes in the resistivity distribution during the irrigation events were related to water movement. The propagation of resistivity change during irrigation events differed for different degrees of soil cracking. For surface cracks of <2 mm width before the irrigation, the resistivity change propagated evenly down-gradient, indicating matrix flow. During irrigation on soil with 30-mm-wide surface cracks, the resistivity change first occurred in the lower parts of the profile before propagating to the top. This suggests preferential flow filling cracks from the bottom up. The differences in initial soil moisture that resulted in these two flow behaviours were reflected in the pre-irrigation resistivity profile. Subsurface temperature changes during the irrigation confirmed the different flow behaviour. 3D Cross-hole ERT allows monitoring of percolation patterns as well as the pre-irrigation moisture states that cause these patterns. This makes 3D cross-hole ERT an excellent tool for researching irrigation management.

Additional keywords: electrical resistivity, preferential flow, soil moisture, 3D tomography.

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Introduction

Fine-grained soils that crack when dry are common in irrigated areas in many parts of the world. While their high nutrient content and water-holding capacity offer favourable conditions for agriculture, the formation of shrinkage cracks during drying intensifies the consequences of suboptimal irrigation scheduling. Irrigation water entering deep cracks can quickly move past the root-zone without being utilised by the plants (Bouma and Dekker 1978; Bronswijk 1991). This water is a waste from an agricultural perspective and contributes to deep drainage, which is often a problem from a hydrological perspective. To assess and improve the efficiency of present irrigation practices, reliable methods to monitor water entry into the soil are needed. Conventional soil moisture measurements are either destructive or provide a moisture reading only for the disturbed vicinity of the instrument. Neutron moisture meters (NMM) (Gardner and Kirkham 1952), time domain reflectometry (TDR) probes (Topp *et al.* 1980), and capacitance probes (Bell *et al.* 1987) all make one-dimensional (1D) measurements. Commercially available TDR and capacitance probes have a relatively small measurement area with a radius of 2–4 cm around the probes, which makes them sensitive to soil disturbance occurring during the installation (Charlesworth 2005). Furthermore, the soil moisture content in the immediate vicinity of the probes might not be representative of natural conditions, as it is often

altered in soil structure and water flow (Rothe *et al.* 1997). Electrical resistivity tomography (ERT) on the other hand allows two-dimensional (2D) and three-dimensional (3D) mapping of soil moisture within undisturbed soil. Even though data are commonly collected with linear surface arrays, theoretically any arrangement of surface and subsurface electrodes can be used. This allows the user to alter the resolution and measurement area based on the specific objectives of the survey. Once the electrodes are installed, data collection requires low labour input and can be automated.

Using a linear array of surface electrodes, Acworth *et al.* (2005) found that bulk soil resistivity of a black Vertosol had a significant correlation with neutron probe readings, indicating that bulk resistivity can provide a valuable measure of soil moisture content in these shrinking and swelling soils. Surface resistivity measurements have successfully been used to monitor percolation in 2D during irrigation events in the field (Michot *et al.* 2003; Batlle-Aguilar *et al.* 2009; Kelly *et al.* 2011). However, Batlle-Aguilar *et al.* (2009) found that the use of a 2D dataset and inversion routine during the monitoring of 3D processes caused underestimation of the depth of the wetting front. To avoid such underestimation and to circumvent the loss of image resolution at depth associated with surface measurements, 3D measurements with subsurface electrodes are necessary.

To measure the soil electrical resistivity in the field, a low-frequency alternating current is applied to the ground and the potential difference between two points is measured. The current (I) is applied by two electrodes (C_1 and C_2), and the resulting voltage (ΔV) is measured between a second pair of electrodes (P_1 and P_2). The apparent electrical resistivity (ρ_a) is obtained by multiplying the resistance ($r = \Delta V/I$) by a geometric factor (K), which accounts for the relative distances of the C and P electrodes, both to each other and to the ground surface. The apparent resistivity that is measured with the four electrodes is a weighted average of the electrical resistivities in the region through which the current travels. To derive the true bulk resistivity from the measured apparent resistivities, an inversion routine is used. Inversion of the field-measured apparent resistivities to infer the 'true' resistivity of the earth is non-unique. This arises from the fact that an infinite number of resistivity distributions can yield the same ρ_a . To limit non-uniqueness, many overlapping ρ_a measurements are taken and inverted simultaneously. By sequentially applying currents and measuring potentials at different locations and over different distances, a range of ρ_a values are measured, which reflect the variation of the electrical resistivity in the subsurface. To estimate the true resistivity distribution in the subsurface, these multiple ρ_a values are then inverted using an iterative inversion and forward modelling process. During this process, a further reduction of the non-uniqueness is achieved through the use of inversion constraints, such as the smoothness constraint in Occam's inversion (deGroot-Hedlin and Constable 1990), which finds the smoothest possible solution that fits the data within a predefined error threshold.

This study describes the use of a novel 3D ERT electrode arrangement that is easily installed in the field and allows 3D measurements with a constant depth resolution. Measurements are made between electrodes that are located in four vertical holes in the corners of a surface square (Fig. 1). The use of subsurface electrodes allows a constant depth resolution, and the arrangement of the multiple electrodes in four vertical holes facilitates electrode installation compared with alternative subsurface electrode arrangements (Garré *et al.* 2011). It is important to note that the imaged volume in this arrangement is located in the undisturbed region between the four probes where the plant root system is developing.

To carry out 3D cross-hole ERT with this electrode layout, automated measurement techniques are required. Automated

resistivity equipment allows measurements to be pre-programmed with any combination of electrodes. An ideal measurement routine would measure all independent electrode configurations (Xu and Noel 1993) at one moment in time. The next-best option is a realistic compromise between acquisition time and an optimised selection of electrode combinations. In addition to an optimised measurement routine, successful ERT needs to consider the measurement scale and measurement environment. For small-scale studies in cracking clay soils as in this study, one needs to consider how to achieve reliable electrode contact in the shrinking and swelling soils, while at the same time maintaining a ratio between the electrode size and the electrode spacing that is small enough to justify the assumption of point current sources during the inversion routine (Loke *et al.* 2003). In addition, electrode polarisation errors are a problem in fine-grained silts and clays that have significant cation exchange capacity (Dahlin 2000). To avoid polarisation errors, the sequence of selected electrode configurations employed during a measurement cycle needs to ensure that polarisation of a current electrode has sufficiently decayed before the electrode is reused as a potential electrode. Most fertile soils used for agricultural production are high in clay and silt content. Therefore, this problem is widely encountered (but frequently not recognised) in the use of electrical resistivity techniques for agricultural investigations.

Details of the challenges for small-scale ERT in cracking soils and how these were accounted for during the design of the electrode strings and the measurement routines are given in Greve (2009). The final design of the electrode strings and the measurement routines are briefly described below.

Methods

3D ERT electrode strings

Electrode strings that provide good electrical contact in cracking clay soils were designed and constructed, and two measurement routines for cross-hole measurements with the selected electrode layout (Fig. 1) were programmed. Four electrode strings each consisting of 16 electrodes with an inter-electrode spacing of 75 mm were constructed. The ring-shaped electrodes (diameter 38 mm) were made from stainless steel grade 316, which provides a good balance between corrosion resistance in the highly reactive clay soils and material-related measurement error

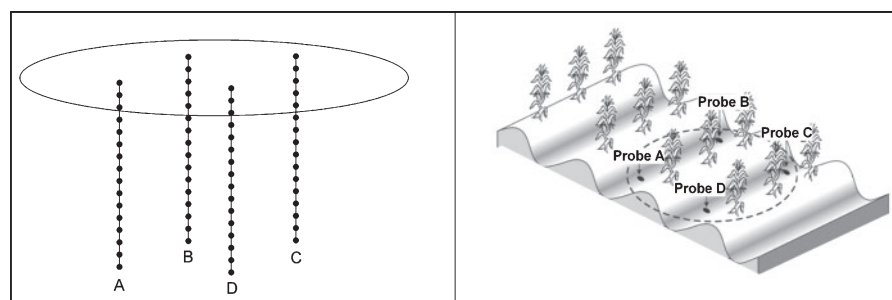


Fig. 1. Left: Subsurface electrode arrangement to carry out 3D cross-hole ERT. Electrode strings are located in four vertical holes (A, B, C, D) that are installed in the corners of a surface square. Right: Schematic of field installation of the probes.

(LaBrecque and Daily 2008). The diameter of the electrodes exceeded the diameter of the remainder of the electrode string by 16 mm (Fig. 2a). This facilitated electrical contact with the soil, as the electrodes stayed in contact with the soil even during soil shrinking when the soil pulled away from the strings.

3D ERT measurement routine

The electrode arrangements for the C and P electrodes in this study were limited to cross-hole measurements between the four electrode strings. For this electrode arrangement, there are six possible measuring planes between the four corners of the measurement square (electrode strings A, B, C, and D in Fig. 1). As recommended by Bing and Greenhalgh (1997, 2000), dipole–dipole measurements with C1 and P1 on one electrode string and C2 and P2 on the other were chosen. To facilitate the selection of suitable measurement routines, the sensitivity distribution within a homogeneous subsurface was simulated for all 1282 possible dipole–dipole measurement combinations between two electrode strings. Two selections of electrode arrangements were then made, a short one for measurements during the irrigation event when resistivity changes occurred rapidly, and a long selection for higher resolution surveys during stable resistivity conditions. Selection of the electrode arrangements was based on achieving measurements with a high and evenly distributed sensitivity. The selected electrode configurations were arranged to allow at least 30 s between reuse of a current electrode as a potential electrode. Testing under field conditions had shown that this interval allowed sufficient decay of electrode polarisation not to interfere with the potential measurement (Greve 2009). The final two measurement protocols consisted of 216 and 2784 electrode arrangements with acquisition times of 3.5 and 45 min, respectively.

3D ERT field installation

The electrode strings were installed in a grey Vertosol within a sorghum field of a commercial farm 25 km south-east of Narrabri, NSW (Fig. 3). Probes were installed halfway between the head and the tail ditch of a 500-m-long field.

Grain size determination of six soil samples at the measurement location gave a soil composition of $45 \pm 5\%$ clay, $20 \pm 1\%$ silt, and $35 \pm 5\%$ sand. The electrode strings were centred on a plant row, with the sides of the surface square orthogonal and parallel to the rows (Fig. 1). A measurement square with a side length of 0.55 m was selected for installation in the field, with the first and the last electrode of each string located at 30 mm and 1155 m, respectively. This spacing provided a satisfactory compromise between sensitivity loss for larger string spacings and a violation of the point assumption during the inversion routine for smaller spacings. The holes for probe installation in the field were augered with a hand-held 40-mm-diameter spiral auger. To assure vertical augering, the auger was held in place by a levelled tripod guide (Fig. 2b). The augered material was collected and worked into a putty-textured soil–water paste, which was used to cover the probes before the installation to ensure a tight fit of the probes in the augered holes (Fig. 2c). Installation of the resistivity probes took place on 1 November 2007, 20 days after plant germination.

Irrigation events

In the growing season 2007–08, the sorghum at the field site was irrigated twice. Alternate furrow irrigation was used, where water is applied to every second furrow during an irrigation event and the furrows that receive water are alternated between consecutive irrigations. Ten rows were irrigated at a time. Irrigation of the rows containing the resistivity probes took place on 15 November 2007 from 07:00 to 19:00 hours and on 2 December 2007 from 10:30 to 22:30 hours. The groundwater that was used for irrigation had an electrical conductivity of 0.45 dS/m. Even application of the entire pumped water during an irrigation event, not taking into account channel water losses, would have resulted in a water application of 100 mm. However, observations made during irrigation showed that the water was not applied evenly between the head and tail ditch and that water was lost to tail runoff. In addition to these two irrigation events, a third watering took place on the 15 January 2008, 2 weeks before the sorghum

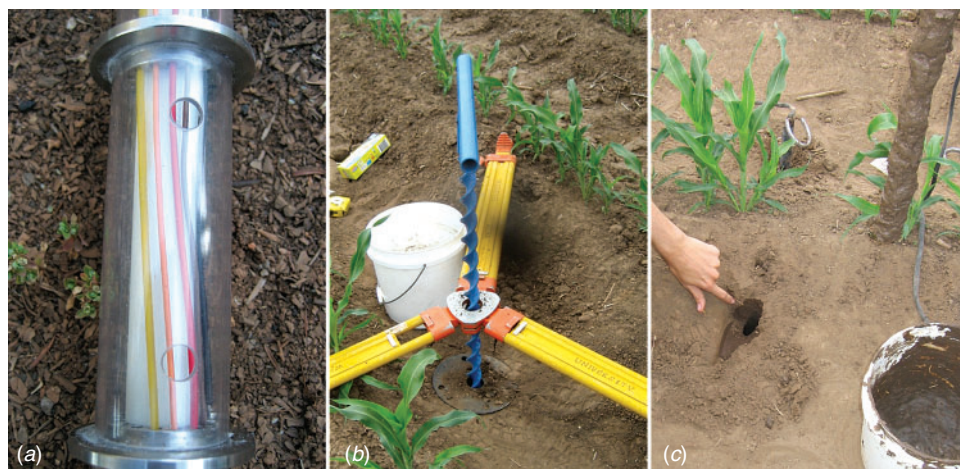


Fig. 2. (a) Section of resistivity probe before filling with silicon; (b) tripod guide for soil augering; (c) resistivity probe covered in soil water paste before insertion.

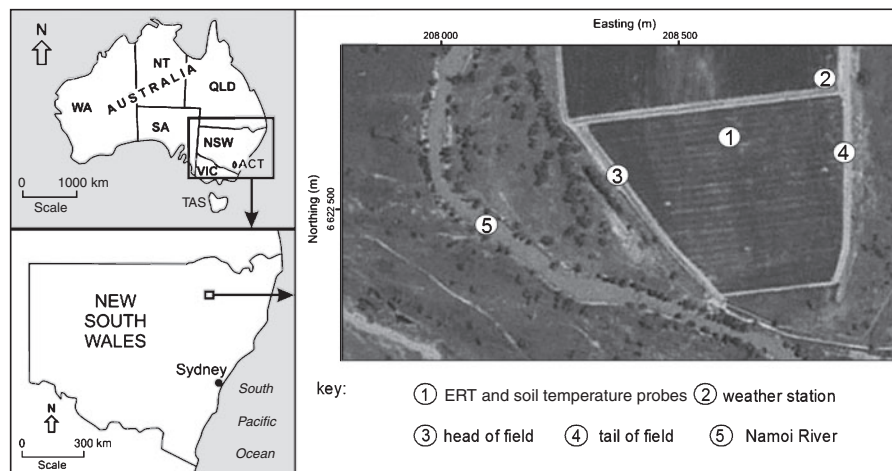


Fig. 3. Fieldsite 25 km south-east of Narrabri, NSW, adjacent to the Namoi River.

was harvested. Surface cracks up to 30 mm wide indicated significantly drier soil conditions at the onset of this watering than before the previous two irrigations, when surface cracks did not exceed a width of 2 mm. For this third irrigation, a two-row-wide application area was confined with two soil walls, which were located 5 m up- and down-gradient of the electrode strings. From 16:20 to 19:00 hours, 2200 L of water was pumped from the side ditch of the field onto the 20-m² area around the probes. The applied water had been pumped from the ground 3 days before the application and hence had an increased temperature compared with the freshly pumped ground water for irrigations 1 and 2. Due to the cracked soil condition, the applied water did not entirely remain in the confined area, as some water was lost into the cracks and reached soil outside of the confinement. To minimise water losses into the cracks, the first 200 L was pumped at a rate of 200 L/h before the pumping rate was increased to 1200 L/h for the remaining 2000 L. Uniform application of the entire volume of water would have resulted in a water application of 110 mm within the confined area.

3D ERT measurements

During an irrigation event, the short protocol (3.5 min) was run every 20 min using a time-lapse setting on the instrument. Once the water flow in the furrows ceased, it was run every 2 h for the first 24 h after the irrigation and every 4 h for the next 24 h. The long protocol was run immediately before the irrigation as well as 24 and 48 h after water flow in the furrows ceased.

Soil temperature measurements

To investigate the downward movement of irrigation water and to allow a temperature correction of the measured resistivities, soil temperature was measured every 15 min at 150, 300, 450, 850, and 1200 mm depth using a probe comprising five individual HOBO Water Temp Pro v2 sensors (Onset Computer Corporation, Bourne, MA). The probe with the temperature sensors was located 2 m up-gradient of the resistivity probes in the same crest as the probes. A weather station was located at the field site, which recorded rainfall using a tipping bucket gauge as well as air temperature and other climate variables (Fig. 3).

3D ERT data processing

The collected 3D ρ_a data were inverted with Res3Dinvx64, ver. 3.02.07 (Geotomo Software, Gelugor, Malaysia, www.geoelectrical.com) with an initial damping factor of 0.15 in a constant depth layer model using the Gauss–Newton optimisation method (Loke 2008). As recommended for time-lapse measurements (Miller *et al.* 2008), measurements with a stacked error of >3% (based on six stacks) were excluded from the inversion. Errors of >3% occurred only in the dry soil before irrigation 3. Exclusion of these measurements led to the rejection of 40 out of 2784 measurements for the long protocol and four out of 216 measurements for the short protocol.

After completion of the inversion, the resistivity was temperature-corrected to account for the daily and seasonal soil temperature changes. A 2% decrease in resistance with every degree increase in temperature was assumed (Hayley *et al.* 2007), and the data were corrected to a standard temperature of 25°C. The correction was based on a linear approximation of soil temperature for the depth of each model layer utilising the measured soil temperature at 150, 300, 450, 850, and 1200 mm as well as the air temperature measured at the weather station.

Results and discussion

The resistivity changes observed during irrigations 1 and 2 were very similar. For brevity the following discussion focuses on irrigations 2 and 3.

The ERT time-lapse series taken during irrigations 2 and 3 show different patterns of resistivity change. As the observed resistivity change occurred throughout an irrigation event, it is safe to conclude that it was caused by a change in soil moisture and possibly by a consequent change in soil structure (e.g. the closure of a shrinkage crack).

During irrigation 2 (Fig. 4), the resistivity first changed at the top of the profile. The initial change occurred under the irrigated furrow before spreading through the crest towards the dry furrow. About 4 h after the irrigation water arrived at the measurement location (bottom row, second left in Fig. 4), the top layer show a homogenous resistivity distribution, indicating that the water from the irrigated furrow had homogeneously

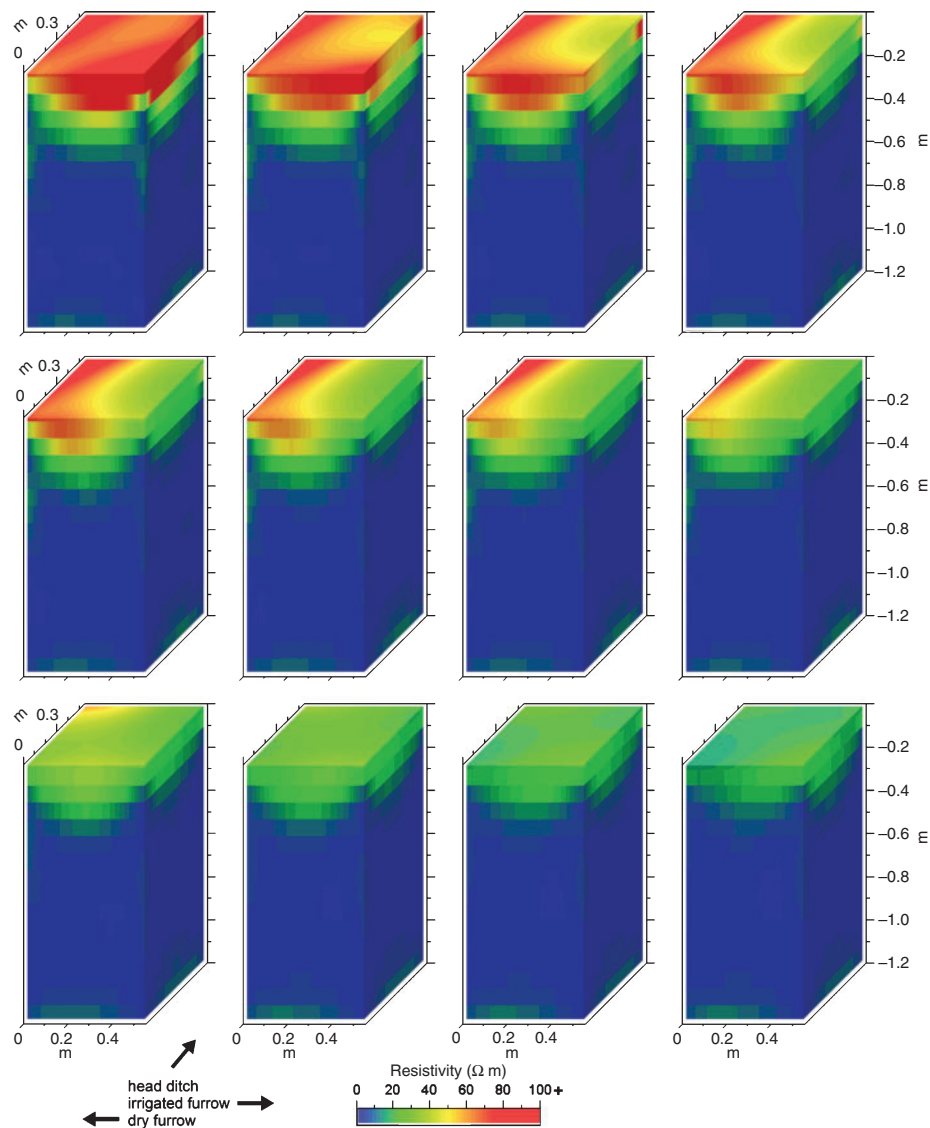


Fig. 4. Irrigation 2: 30-min steps of a time-lapse series throughout the irrigation taken with the short protocol. The first image (upper left) was taken 30 min before the irrigation water arrived at the measurement location.

wetted the crest and the dry furrow. This downward and sideways propagation of the resistivity change during irrigation 2 indicates water movement dominated by matrix flow.

The propagation of resistivity change during irrigation 3 (Fig. 5) indicates a different flow behaviour. Resistivity changes were first detected in the dry zones at 1 m depth followed by an upward propagation of resistivity change throughout the irrigation event. This upward propagation indicates that water flowed into the soil cracks that were observed at the onset of this irrigation, resulting in crack filling and crack closure from the bottom up. Such behaviour is typical for an ‘internal catchment’ in cracked soil, as described by Van Stiphout *et al.* (1987).

The soil at the top of the profile would have taken up some of the water that was flowing past. However, this soil moisture

change was not detected, as the presence of the cracks in the soil dominated the soil’s resistivity response. The bulk resistivity at the top of the profile was only reduced once the cracks at this depth were closed, or at least filled with water.

The high resolution ERT images that were taken before and after the irrigation process provide an insight into the soil moisture and soil cracking state that resulted in these two different percolation behaviours. The 3D profile collected before irrigation 3 showed significantly higher resistivities than the profile collected before irrigation 2 (Fig. 6a, b). This highlights the drier soil conditions before irrigation 3, which resulted in the observed preferential flow.

As the typical resistivity of clays ranges from ~ 2 to $100 \Omega\text{m}$ (Sharma 1997), resistivity changes of $>100 \Omega\text{m}$ during an irrigation event on clay soils can be attributed to structural change, such as the closure of shrinkage cracks (Greve *et al.*

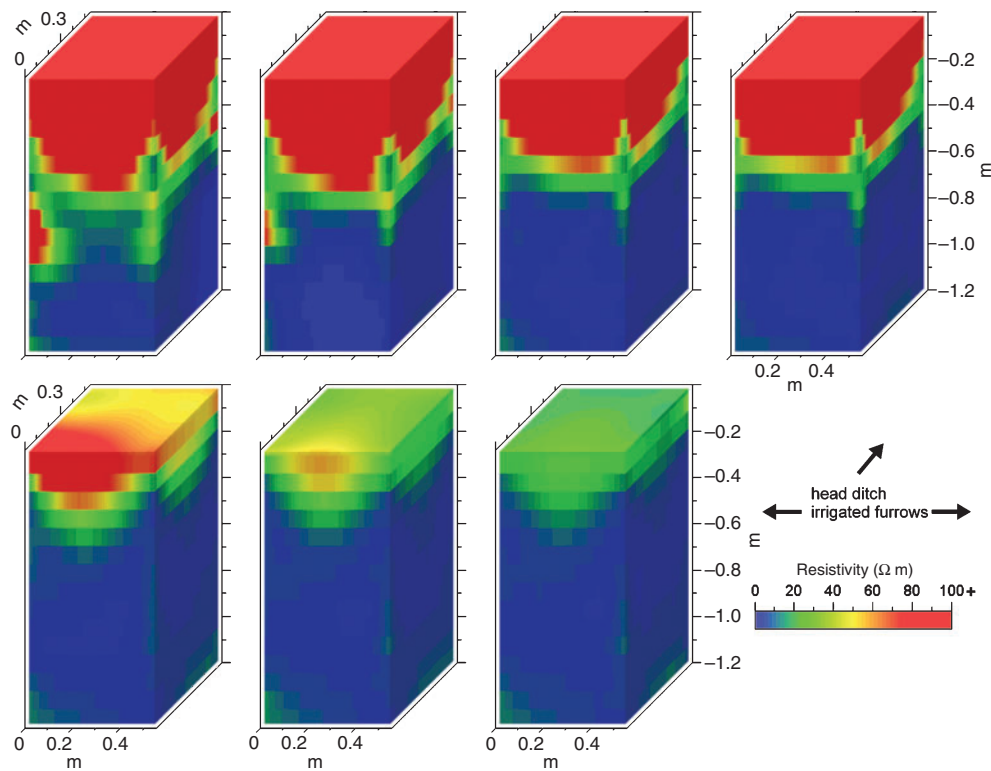


Fig. 5. Irrigation 3: 30-min steps of a time-lapse series throughout the irrigation taken with the short protocol. Irrigation of the measurement area commenced immediately after the first image (upper left) was taken.

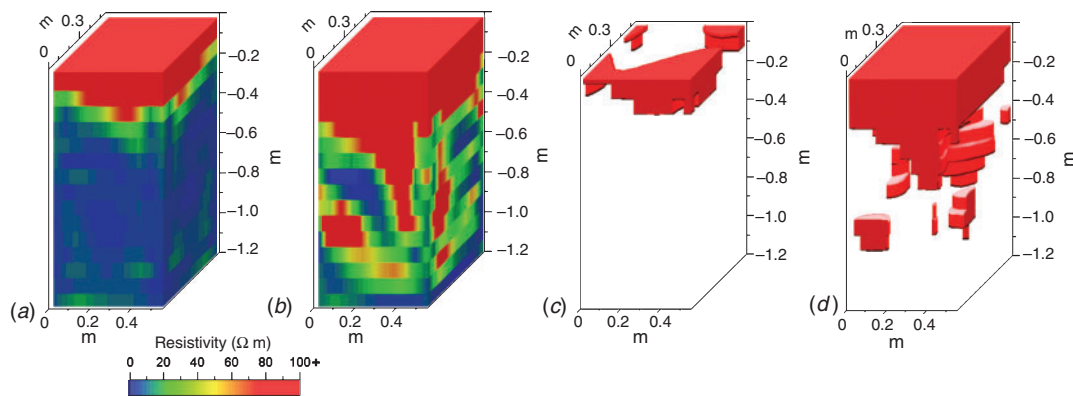


Fig. 6. Resistivity data collected with the long protocol; (a, b) Resistivity distribution before irrigation 2 and 3, respectively; (c, d) areas of resistivity image that went through a change of $>100 \Omega\text{m}$.

2010). Figure 6c, d shows the areas that had a resistivity change of $>100 \Omega\text{m}$ during irrigations 2 and 3. The smoothness constraint used during the inversion routine causes a smearing of sharp resistivity contrasts as those that would occur along a crack wall (Dey and Morrison 1979). Therefore, Fig. 6 does not show the individual cracks, but rather gives an indication of cracking depth and crack spacing. Classical crack theory states that different crack depths and spacings in the field result in several small surface soil prisms being joined together at their base to form larger prisms in the subsurface (White 1970). The shape of the crack-affected area in the dry soil before irrigation 3 reflects this crack

geometry. The finer spaced cracks of up to 300 mm depth cannot be separated in the resistivity image and appear as a continuous crack-affected area. At greater depth, the crack spacing is increased and the crack-affected area forms elongated, isolated regions to a depth of 1000 mm. The isolated, crack-affected areas between 700 and 900 mm depth in Fig. 6 are located at the boarder of the modelling area and are therefore most likely the result of a crack diverting from the vertical and entering the modelling area at this depth. Before irrigation 2, the volume and depth of the crack-affected area is significantly smaller than before irrigation 3. The crack affected area only reaches a depth of 300 mm and the crack spacing in this upper

300 mm is wide enough to be detected as separate cracking areas by the inversion routine.

The different extent of the crack-affected area before the two irrigation events is consistent with the different percolation behaviour highlighted in the time-lapse series in Figs 4 and 5. The smaller crack-affected area before irrigation 2 is associated with matrix-dominated flow and the larger crack-affected area before irrigation 3 is associated with preferential flow.

The soil temperature changes that were measured next to the resistivity probes (Fig. 7) provide an independent check of the information given by the 3D ERT profiles. During irrigation 2, the diurnal temperature fluctuations continued throughout the irrigation process, and only at the very top of the soil profile a dampening of the soil temperature occurred. The water for irrigation 2 had been freshly pumped from the ground and had a temperature of 21°C, which would have caused the observed cooling of the upper soil layers. Water for irrigation 3, on the other hand, had been pumped from the ground 3 days before application and had a temperature of 33°C. During the application of this water, a sudden increase in soil temperature occurred at the measurement depths from 300 to 1200 mm (Fig. 7). The lack of temperature response at the very top of the soil profile during this irrigation can be explained by the lack of temperature gradient between the applied water and the soil at this depth. The temperature increase in the rest of the profile was detected at different logging intervals. At 850 and 1200 mm, the increase was detected at 17:30 hours, while the temperature increase at 300 and 450 mm was detected during the next logging interval at 17:45 hours. The timing of these temperature rises coincided with the increase in the water application rate at 17:20 hours. When the water application rate was increased, water ponding and water flow into the soil cracks was observed. The temperature rise can thus be attributed to advective heat transport caused by water flowing into the soil cracks. The timing and depth of these temperature changes provides a valuable independent check of the electrical

resistivity interpretation. The fact that the temperature at 1200 and 850 mm depth increased before the temperature increase at 450 and 300 mm shows that preferential flow caused crack filling from the bottom up, just as indicated in the resistivity time-lapse series in Fig. 5. The temperature change at 850 mm was significantly larger than at 1200 mm. This indicates that less water and therefore less heat arrived at 1200 mm, suggesting that the crack volume decreased between 850 and 1200 mm depth. The decrease in crack volume between 850 and 1200 mm depth indicated by the temperature measurements confirms the extent of the crack-affected area in Fig. 6 as derived based on the ERT profiles. And the lack of a sharp temperature response during irrigation 2 (Fig. 7) confirms the results of the ERT profiles, which show matrix-dominated flow and no deep-reaching cracks in the profile.

Using temperature changes in the subsurface to investigate flow behaviour depends on a sufficiently large temperature contrast between the soil and the applied water. In contrast to 3D ERT, it is therefore not suitable for routine use during irrigation management.

Of most interest from an irrigation management perspective is the determination of the extent of cracking needed to cause a transition from matrix-dominated flow to preferential flow and the depth of the potential preferential flow paths. Even though some deep drainage might be favourable during irrigation management as it leaches accumulated salts out of the root-zone (Vervoort *et al.* 2003), fast deep drainage via preferential flow paths is ineffective in removing salts. An irrigator would therefore want to avoid a cracking extent that results in fast preferential flow past the root-zone. To do so, information about the expected flow behaviour and the depth of the potential preferential flow paths is needed before an irrigation event.

Continued application of 3D ERT with the long and the short protocol before and during irrigation events will allow an irrigator to move from monitoring different percolation patterns to identifying the pre-irrigation resistivity profiles that will result

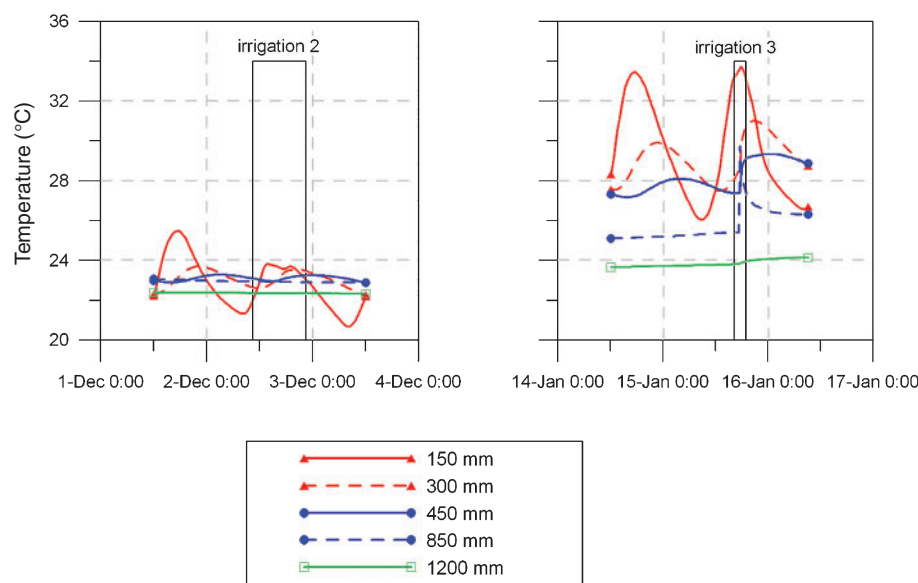


Fig. 7. Soil temperature at different depth during irrigation 2 and irrigation 3.

in these patterns. This knowledge will allow optimised irrigation scheduling, which aims at minimising the number of irrigations without compromising plant growth or creating unfavourable preferential flow conditions.

Summary and conclusions

3D Cross-hole ERT has been carried out during three irrigation events in a sorghum field. Irrigations were carried out on soil with different surface crack intensities, and changes in the resistivity distribution during the irrigation events were related to water percolation processes. The propagation of resistivity change during water infiltration was found to differ for different degrees of soil cracking. When surface cracks of <2 mm width were observed before the onset of the irrigation, the resistivity change propagated evenly down gradient, indicating matrix flow. However, during irrigation on drier soil with surface cracks of up to 30 mm width, the resistivity change was first detected in the lower parts of the profile before propagating to the top of the modelling area. This highlights the presence of an internal catchment, resulting in preferential flow filling the soil cracks from the bottom up. Differences in the soil moisture and soil cracking state, resulting in these two percolation behaviours, were clearly reflected in the high resolution, pre-irrigation resistivity profiles. The crack-affected area before irrigation 3 extended to a depth of 1000 mm, whereas before irrigation 2, it was only 300 mm deep.

The different percolation behaviours highlighted in the ERT profiles are also shown by the soil temperature changes measured during water application. In soil conditions with surface cracks <2 mm wide, the diurnal temperature variations in the soil profile continued throughout the water application, and only at the upper measurement point at 150 mm was a moderate dampening of the diurnal temperature signal detected. This is in contrast to the temperature changes observed during irrigation 3, when the soil profile contained extensive cracks. During this irrigation, a sharp increase in temperature was observed throughout the measuring profile. The temperature increase was first detected at the lower two measuring depths (850 mm and 1200 mm) before it was sensed further up in the profile. This upward propagation of temperature change confirms the presence of an internal catchment in the cracked soil with preferential flow, filling soil cracks from the bottom up.

Effective irrigation scheduling aims at minimising the number of irrigations while not compromising plant growth. During an irrigation event, the applied water ought to evenly wet the root-zone while minimising deep drainage and surface runoff. Low application rates resulting in crack closure before the entire profile is wetted favour the occurrence of surface runoff (Mitchell and Van Genuchten 1993), and high application rates that cause water ponding in deep cracks could result in water moving past the root-zone (Bouma and Dekker 1978). Determining the optimal application rate and duration requires a good understanding of soil moisture state and flow processes within the root-zone. This requires information about the expected percolation behaviour and the depth of the potential preferential flow paths before an irrigation event. 3D ERT as presented in this study allows monitoring of percolation patterns and detection of the cracking extents that result in these patterns.

Continued application of this method before and during irrigation events will help to advance from simply monitoring these flow patterns to identifying the pre-irrigation resistivity profiles that will cause them.

Further improvement of 3D ERT as an irrigation management tool could be achieved through resistivity calibration relationships that allow quantifying soil moisture in cracking soils based on ERT profiles.

Acknowledgments

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